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Morpho Software Inc.

Executive Summary

Introduction

There are more than 1.5 billion Web pages today and it is estimated to reach 8 billion Web pages by 2003. Furthermore, higher network bandwidth, higher processing speed, and larger and cheaper storage are driving the text-based Web to become more and more multimedia-based. These trends result in explosive image and video Web content, which demands Web tools capable of effectively and efficiently *organizing*, *filtering*, and *retrieving* multimedia data.

The mission of Morpho Software is to develop multimedia-enabling technologies to *morph* today's Web into the future. Specifically, Morpho Software offers ***Perception-based Multimedia Information Filtering and Retrieval Software***. Initial products/services include an Objectionable Image Filter and a Perception-based Image Retrieval (PBIR) Engine.

Morpho Software was founded in February 2001 by scientists from Univ. of California, Santa Barbara and Stanford who have extensive experience in analysis, indexing and retrieval of multimedia information. Morpho Software has innovative and patent-pending technologies that support personalized multimedia information filtering and retrieval.

Products/Services

A. **Objectionable Image Filter:** Filtering software is increasingly needed to screen out objectionable images from minors. Our **M**ulti-model **O**bjectionable info**R**mation **F**ilter (MORF) can perform effective real time filtering of pornographic images. In addition, MORF can be personalized to meet different filtering criteria. This software is ideal for ISP to ensure an adequate level of protection for minors without undue cost, inconvenience or interference.

   MORF is superior to existing image filtering software, which uses either text or URL to determine an objectionable site and has encountered various limitations.

B. **Perception-Based Image Search Engine:** Our **P**erception-**B**ased **I**mage **R**etrieval (PBIR) engine helps users of image content providers find the desired images much faster and much more easily. Our system can be integrated seamlessly with existing keyword-based image search to substantially shorten users' search time and effort.

   For image search tasks, a query concept is hard to articulate, and articulation can be subjective. For instance, it is difficult to describe a desired image using low-level features such as color, shape, and texture. Different users may perceive the same image differently; and even if an image is perceived similarly, users may use different vocabulary (i.e., different combinations of low-level features) to depict it. Furthermore, most users are not trained to specify simple query criteria using formal query languages. Morpho's image search engine can quickly learn users' query concepts via relevance

feedback to capture subjective concepts and alleviate users from specifying complex queries for image search. Morphosoft's image retrieval technologies have been recognized by top researchers in the field as "by far the best image search engine that is one generation ahead of any other products in the market."

Founding Members and Technical Advisor

Dr. Edward Y. Chang is Co-founder of Morpho Software.
- –Professor, Electrical and Computer Engineering, Univ. of California, Santa Barbara
- –M.S. UC Berkeley, M.S. Stanford, Ph.D. Stanford
- –10 years industry experience in large-scale software architecture/development (DEC, SUN Microsystems)
- –Executive committee member, Univ. of California Digital Media Innovation Program
- –IBM Faculty Partnership Award

Dr. Kwang-Ting (Tim) Cheng is Co-founder of Morpho Software.
- –Professor, Electrical & Computer Engineering, Univ. of California, Santa Barbara
- –Director, Computer Engineering Program, Univ. of California, Santa Barbara
- –B.S. National Taiwan Univ., PhD. UC Berkeley
- –Chief Technology Advisor, Verplex System Inc.
- –Technical Advisor of 5 Silicon Valley based high-tech companies
- –Consultant of Lucent Technologies, LSI Logic, Institute for Information Industry, Taiwan
- –Fellow of IEEE

Dr. Hector Garcia-Molina is Technical Advisor of Morpho Software.
- –The Leonard Bosack and Sandra Lerner Professor, Departments of Computer Science and Electrical Engineering, Stanford University
- –Chairman, Department of Computer Science, Stanford University, starting Jan. 2001
- –Director, Computer Systems Laboratory, Stanford University from 1991-1994
- –Professor, Computer Science Department, Princeton University from 1979-1991
- –Member, United States President's Information Technology Advisory Committee (PITAC).
- –Fellow of the ACM

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